J&J HR Digit × databricks

WORKSHOP WITH J&J HR & HR DIGITAL

Databricks MLOps & AI

From Raw Data to Deployed AI Agent — End-to-End MLOps on the Databricks Data Intelligence Platform

MLFLOW 3.0 | DELTA LAKEHOUSE | AI AGENT | UNITY CATALOG | DATABRICKS APPS

THE DATABRICKS DATA INTELLIGENCE PLATFORM

Everything J&J HR Digital Needs — One Platform

USER EXPERIENCE

- Databricks Apps
- Databricks One
- MS Teams
- 3rd Party BI Tools

AI AGENT LAYER

- Agent Bricks
- Genie
- AI Gateway · LLMs
- Agent / MCP Registry
- Agent Orchestration · ResponsesAgent · MLflow 3.0 · A2A Protocol

ML & MLOPS

- MLflow 3.0 · Experiments
- Model Registry · UC-native
- Model Serving · REST Endpoints
- Feature Store
- Versioning · Models & Agents
- AutoML · sklearn · XGBoost

GOVERNANCE

- Unity Catalog · Organize / Manage / Secure
- Data Classification · Tags
- Quality Monitors · Freshness · Completeness · Drift Detection
- End-to-End Lineage
- Full Auditability · System Tables

DATA PLATFORM

- Lakeflow · ETL Pipelines
- Lakehouse · Bronze / Silver / Gold
- Volumes · Unstructured Data
- Dashboards
- Business Semantics

END-TO-END MLOPS · AI · GOVERNANCE

The Full Databricks MLOps Platform

From raw data landing to a deployed AI agent — unified, governed, and observable at every step.

BRONZE	SILVER + UC	GOLD + AI	MLFLOW	MLFLOW 3.0	APPS
① Ingest	② Govern	③ Build for AI	④ Train ML	⑤ Agent	⑥ Deploy
Delta Lake	PII · Masking · DQX	Genie · Vector Search	Experiments · Registry	ResponsesAgent · UC	Databricks Apps · Monitor

GOVERNANCE	EXPERIMENTS	REST ENDPOINTS	SEMANTIC RAG	AI APPS
Unity Catalog	MLflow 3.0	Model Serving	Vector Search	Databricks Apps

MLflow 3.0 — unified tracking, evaluation, and governance across every step

MLFLOW 3.0 · END-TO-END

The ML Lifecycle on Databricks

From feature engineering to production monitoring — MLflow 3.0 tracks every experiment, registers every model, and governs every deployment through Unity Catalog.

The ML lifecycle on Databricks — from data to deployment

AI AGENTS

Databricks Agents AI Stack

Three integrated layers — from raw compute to prebuilt production agents — with end-to-end governance across the full agent lifecycle.

Agent Bricks

Prebuilt contextual reasoning agents ready to deploy — Knowledge Assistant, Supervisor Agent, Documents Agent, AI Functions.

→ No build required
→ UC-governed by default
→ Customisable via instructions

Agent Platform

Developer platform for building, evaluating, and serving agents. Agent Orchestration, LLMs, MLflow Tracing & Evals, AI Gateway, Agent/Skill/MCP Registry.

→ A2A · MCP Registry
→ MLflow Tracing & Evals
→ Gemini · Claude · OpenAI

Serverless GPU

On-demand best-in-class accelerated hardware integrated directly into Databricks notebooks and jobs. No infrastructure management — just scale.

→ Fine-tuning at scale
→ BYOM or marketplace
→ Scale to zero

Agent-to-Agent

Multi-turn delegation across any orchestrator

MCP Registry

Auto-discovery for Claude Code, Cursor & any client

Observability

MLflow tracing · spans · SQL-queryable logs

Unity Catalog

RBAC · lineage · audit trail across all agents

MLFLOW 3.0

ResponsesAgent — The New Agent Standard

MLflow 3.0 introduced **ResponsesAgent** — an OpenAI Responses API-compatible interface that turns any framework or model into a first-class, governable Databricks service.

🔌

Drop-in OpenAI Replacement

Any client built for OpenAI works with your Databricks agent — just swap `base_url`. No SDK changes, no prompt rewrites.

🏗️

Wrap Any Framework

LangChain, LlamaIndex, LangGraph, custom Python — wrap it in `ResponsesAgent` and deploy through the same governed interface.

📊

MLflow Tracing + UC Governance

Every agent call is tracked end-to-end with MLflow spans, versioned in the model registry, and subject to Unity Catalog permissions — full enterprise governance on agentic workloads.

⚖️

Quality Gates via `mlflow.genai.evaluate()`

Built-in LLM judges score safety, relevance, and custom criteria before deployment. Results flow to Delta tables — queryable with SQL.

EXPERIMENTS	REGISTRY	OBSERVABILITY	EVALUATION
Run Tracking	**UC Model Registry**	**Agent Tracing**	**LLM Judges**

LIVE

DEMO

Hire Right — Predictive Hiring Agent for J&J HR Digital

QUERY_GENIE SEARCH_RESUMES PREDICT_HIRING_SCORE SEND_EMAIL

END-TO-END SOLUTION

Hire Right Architecture

HR Manager
J&J HR Digital

① chat →

Hire Right App
Databricks App · FastAPI + HTML

② invoke →

Hire Right Agent
ResponsesAgent · MLflow 3.0
OpenAI-compatible

TOOLS

query_genie
Genie Space · NL-to-SQL

search_resumes
Vector Search · RAG

predict_hiring_score
ML Endpoint · sklearn

send_email
Mailgun API · external

Delta Lakehouse
Bronze → Silver → Gold
Candidate Data · Resume PDFs

Unity Catalog
Data Classification · Masking
Data Quality Monitor

MLflow 3.0
Experiments · Model Registry
Agent Tracing · Evaluation

Agent Bricks
AI Gateway · MCP Registry
Foundational LLMs

① · NOTEBOOK 00

Ingest — Bronze Layer

Raw candidate data lands in Delta Lake as a Bronze table — schema inferred from CSV, resumes uploaded as PDFs to a Unity Catalog Volume.

UC Catalog Explorer showing raw_data volume

STEP 1 · WORKFLOW ORCHESTRATION

Databricks Workflow — Data Pipeline DAG

The full end-to-end pipeline is orchestrated as a Databricks Workflow — each notebook runs as a serverless task, with lineage and run history tracked automatically.

Databricks Workflow DAG: Bronze to Gold pipeline tasks

STEP 1 · WORKFLOW ORCHESTRATION

Databricks Workflow — ML & Agent Pipeline

Downstream tasks fan out from the data pipeline — Vector Search index, Genie Space, ML training, Agent registration, and dashboard refresh all run as a single coordinated job.

Databricks Workflow DAG: ML, Agent, Vector Search and monitoring tasks

② · NOTEBOOKS 01-02

Govern & Enrich — Silver Layer

Silver applies business transformations and Unity Catalog governance — PII tagging, column masking, data quality monitoring, and rich AI-ready comments.

PII Tags · email · phone | Column Mask · mask_phone() | sensitivity = internal | Data Quality Monitor | PRIMARY KEY constraint

Unity Catalog candidates table with PII tags and column masking

③ · NOTEBOOKS 03-04

Build for AI — Gold Layer + Genie Space

Gold tables aggregate candidate scores for analytics and ML training. Genie Space gives the agent natural language access to HR data. Vector Search enables semantic resume retrieval.

Candidates table sample data with masked phone numbers

④ · NOTEBOOK 05

Train ML Model — Hire / No-Hire Prediction

MLflow autolog captures every experiment run. Best model (GradientBoostingClassifier) selected by AUC-ROC and registered to Unity Catalog with the `@champion` alias.

MLflow Experiments with multiple model runs

REGISTER · NOTEBOOK 05

Register & Serve — @champion Model Endpoint

Best run registered to Unity Catalog Model Registry and aliased as `@champion`. A serverless Model Serving endpoint exposes it as a REST API — the agent calls it via `predict_hiring_score`.

Model Serving endpoints

⑤ · NOTEBOOK 06

Build & Deploy Agent — ResponsesAgent + MLflow 3.0

The Hire Right Agent is a `ResponsesAgent` (OpenAI-compatible ChatAgent) packaged with MLflow 3.0, evaluated with `mlflow.genai.evaluate()`, and deployed as a serverless endpoint.

MLflow evaluation with JnJ Tone judge scores

⑤ · MLFLOW EVALUATE

Agent Quality Gates — LLM Judges & Feedback

Custom JnJ Tone scorer plus built-in safety and relevance judges run against the agent before deployment. Every assessment is logged to MLflow and queryable with SQL.

MLflow agent evaluation results table

MLflow trace with JnJ Tone assessment detail

⑥ · NOTEBOOK 07

Deploy — Hire Right App on Databricks Apps

The Hire Right App is a Databricks App (FastAPI backend + vanilla HTML/JS frontend) that proxies chat messages to the agent endpoint — with Databricks OAuth handling auth seamlessly.

Databricks Apps interface showing hire-right-agent

Hire Right app showing candidate scoring dashboard

MONITOR · CONTINUOUS

Monitor & Govern — Unity Catalog Throughout

Data quality monitors and model drift detection run continuously. Unity Catalog enforces governance at every layer — from raw Bronze to deployed agent.

Profile Metrics Table | Drift Metrics Table | Column Lineage | PII Audit Log | UC Access Controls

MLflow tracing timeline with tool calls

Databricks Agents registry showing hire-right-agent-endpoint

REFERENCE

Appendix

Detailed capability mapping, charter, and supporting reference material.

CAPABILITY MAPPING

WORKSHOP CHARTER

J&J HR DIGITAL WORKSTREAM CHARTERS

Mapping Charter Goals to Databricks

CHARTER CAPABILITY	DATABRICKS CAPABILITY
Multi-layer HR data platform (Bronze → Silver → Gold)	Delta Lakehouse · Medallion Architecture
Raw ingestion from HRIS systems (ADP, Workday, resumes)	Delta Lake · read_files() · Autoloader · UC Volumes
PII governance, data cleansing & classification	Unity Catalog · Data Classification · Column Masking
Analytics-ready HR data model & semantic layer	AI/BI · Genie Space · Dashboards · UC Comments
Data quality monitoring & observability	Lakehouse Monitors · Profile & Drift Metrics
Foundational MLOps environment — experiment tracking & registry	MLflow 3.0 · Experiments · UC Model Registry
Model deployment automation & CI/CD pipelines	Model Serving · agents.deploy() · Databricks Workflows
Decision services for priority HR use cases	ResponsesAgent · Agent Bricks · Databricks Apps
Model governance, bias auditing & fairness review	AI Gateway · Agent Registry · LLM Judges
Production monitoring, drift detection & retraining	Lakehouse Monitors · MLflow Tracing · Workflows

J&J HR Digital × databricks

Building the Future of HR Intelligence Together

From Bronze ingestion to AI-powered hiring decisions — Databricks delivers the complete Data Intelligence Platform for J&J HR Digital's predictive hiring vision.

8	4	1	∞
MLOps Steps	Agent Tools	Platform	Possibilities